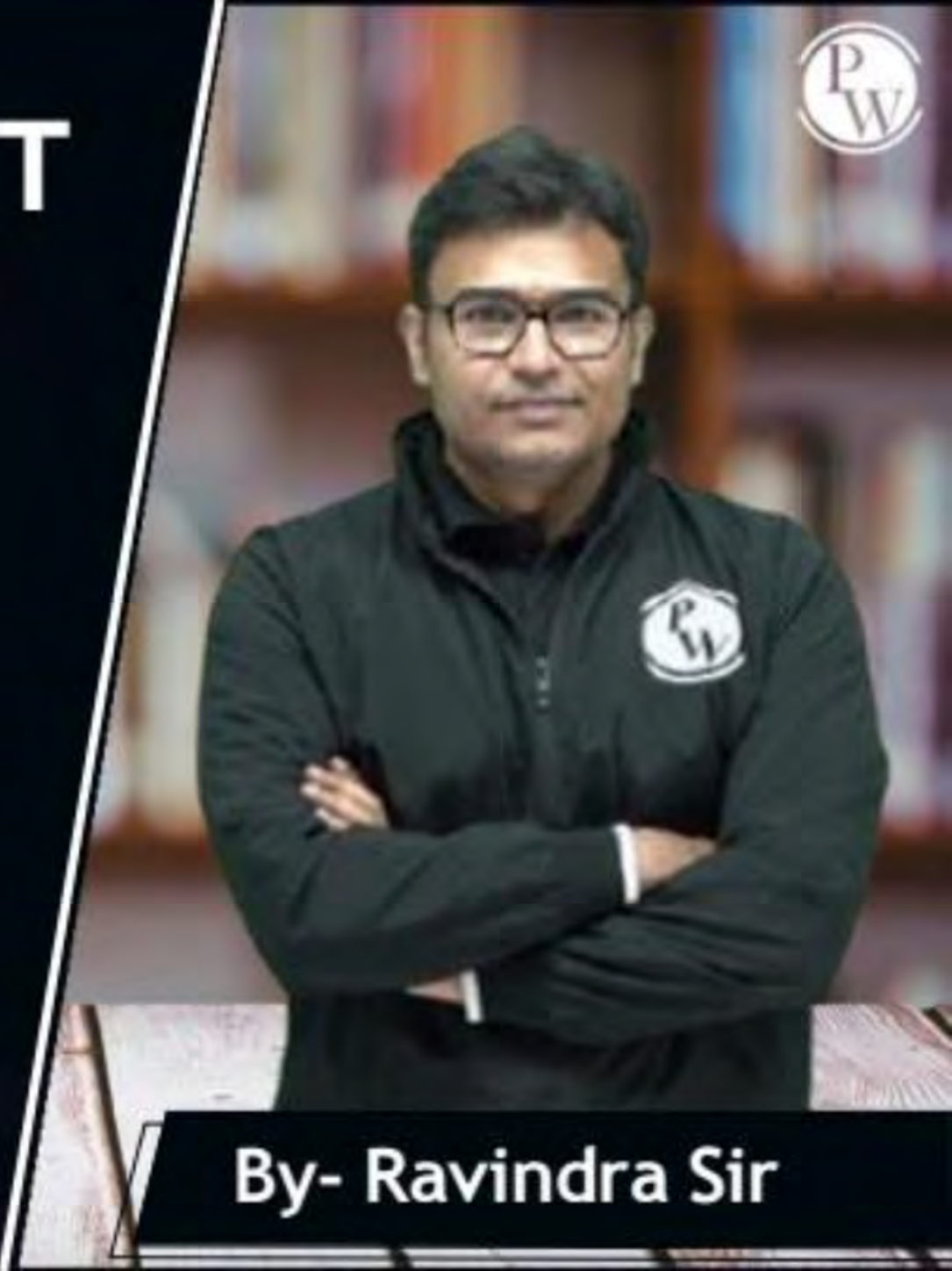
Computer Science & IT

COMPUTER NETWORK (CN)

TCP and UDP

Lecture No. 04



By- Ravindra Sir

Recap of Previous Lecture



Topic Bridges

Topic

Topic

Topics to be Covered



- Topic — Bridges
- Topic
- Topic

To make bridges transparent, two algorithms are used

1) Backward learning algorithm to stop traffic being sent where it is not needed
2) A spanning tree algorithm to break loops that might be formed when bridges are cabled together

Backward learning algorithm :-



Hash table

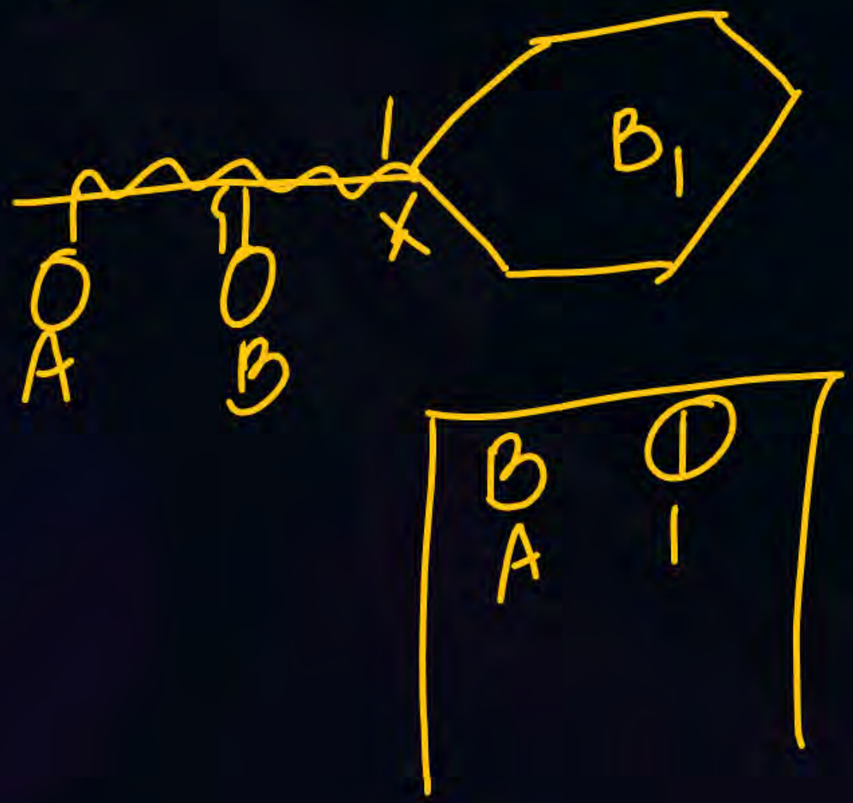
Host	Port	time
A	1	5 min ✓
B	1	" ✓
C	1	" ✓
D	2	" ✓
E	2	" ✓
F	2	" ✓

→ whenever a bridge receives a packet, it will update the hash table

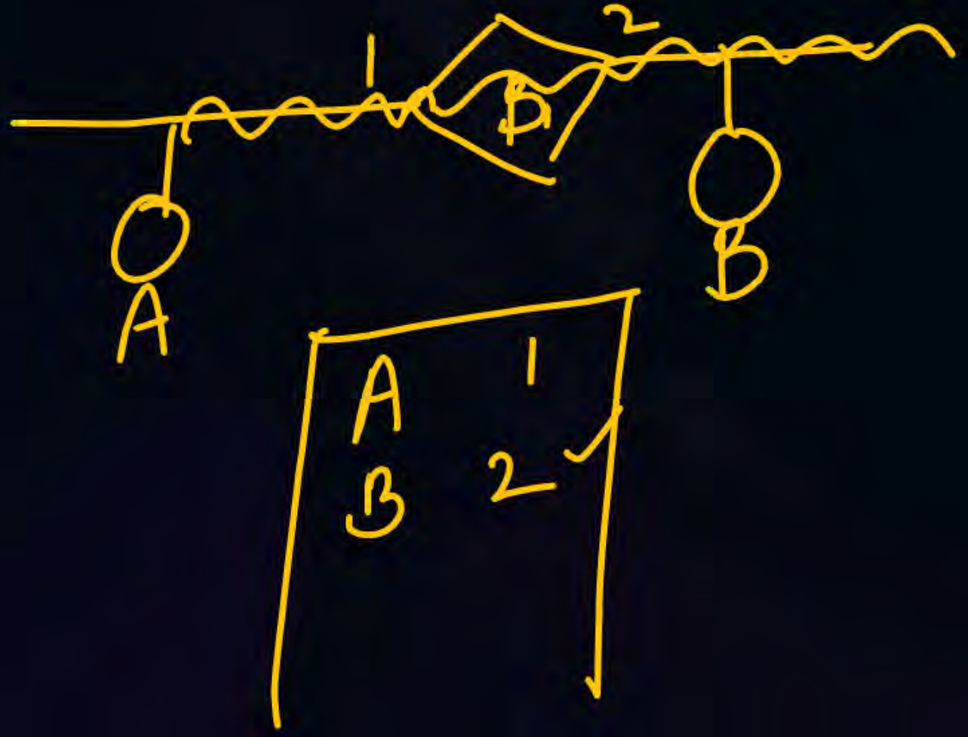
→ Hash table entries are used for small amount of time because host may change their location within a LAN.

The Bridges will do three things :-

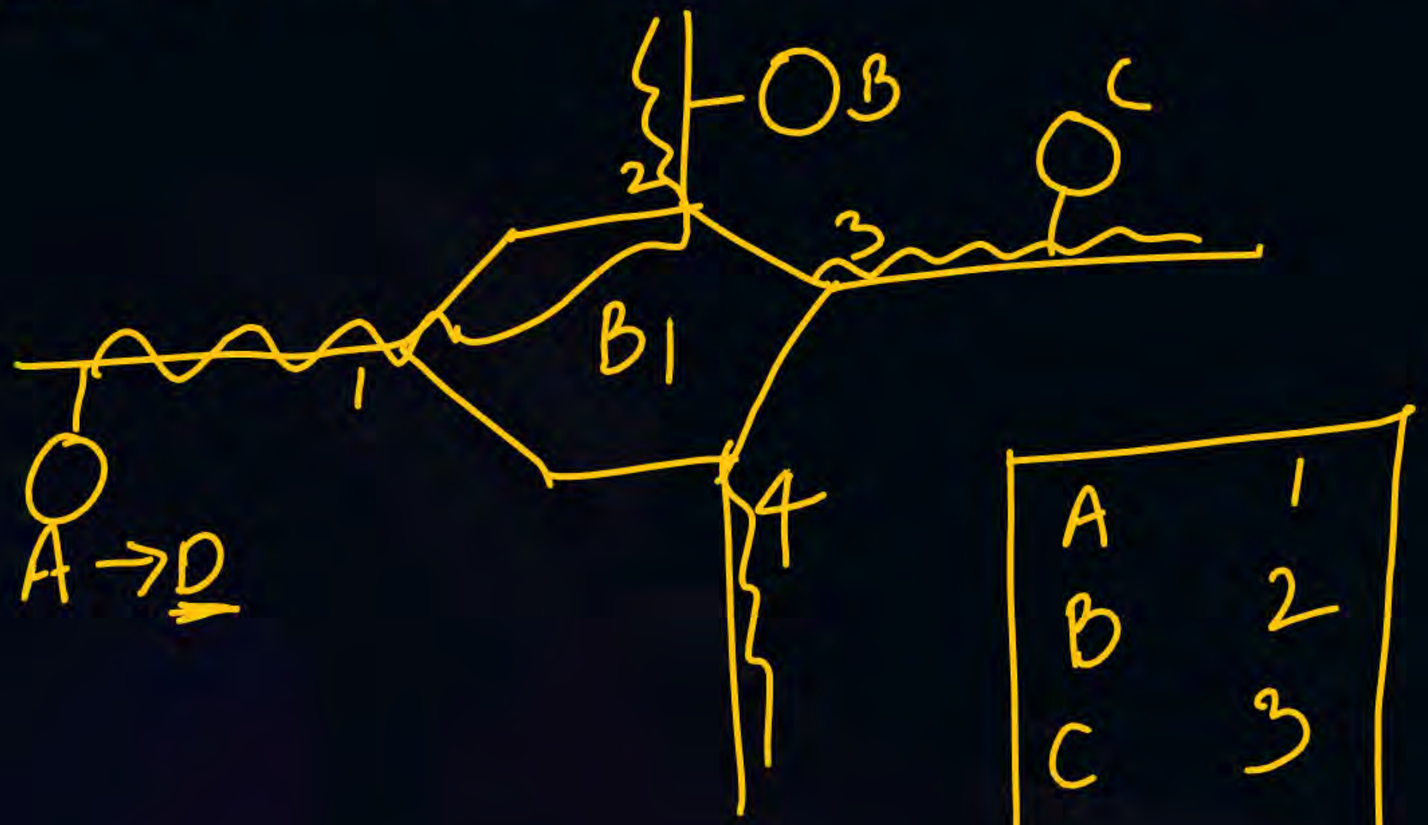
1) If the port for the destination address is same as the port for the source address, it discards the frame



2) If the port for destination address and the Source port are different, forward the frame on destination port



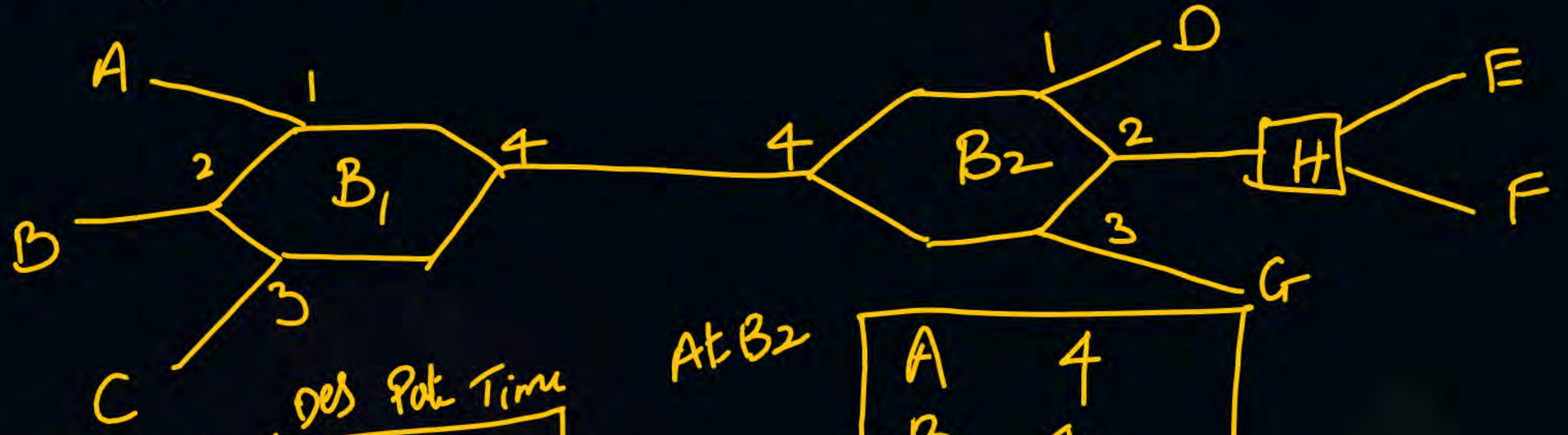
3) If the destination port is unknown, uses flooding and send the frame on all outgoing ports except the source port



A	1
B	2
C	3

1) Forward ✓
2) Filter ✓
3) Flooding ✓

Bridges can use point to point connections



At B_1:

Des	Port	Time
A	1	
B	2	
C	3	
D	4	
E	4	
F	4	
G	4	

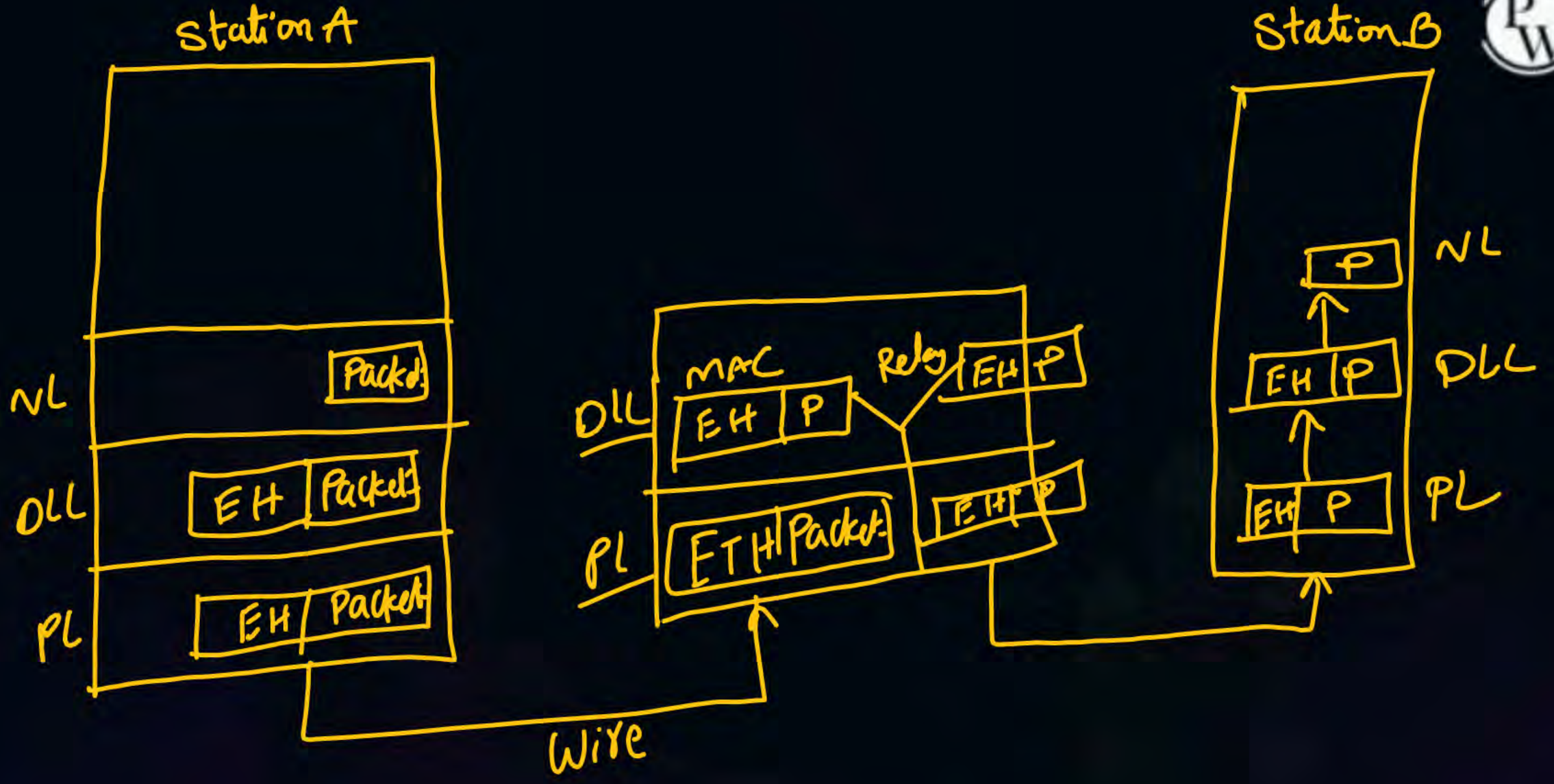
At B_2

A	4
B	4
C	4
D	1
E	2
F	2
G	3

As a frame arrives on a link, the VLSI chips in Bridges will look at the MAC add (PL, DLL → Bridges) and start forwarding it on destination port immediately, even if full packet has not arrived. It is called Cut through Switching

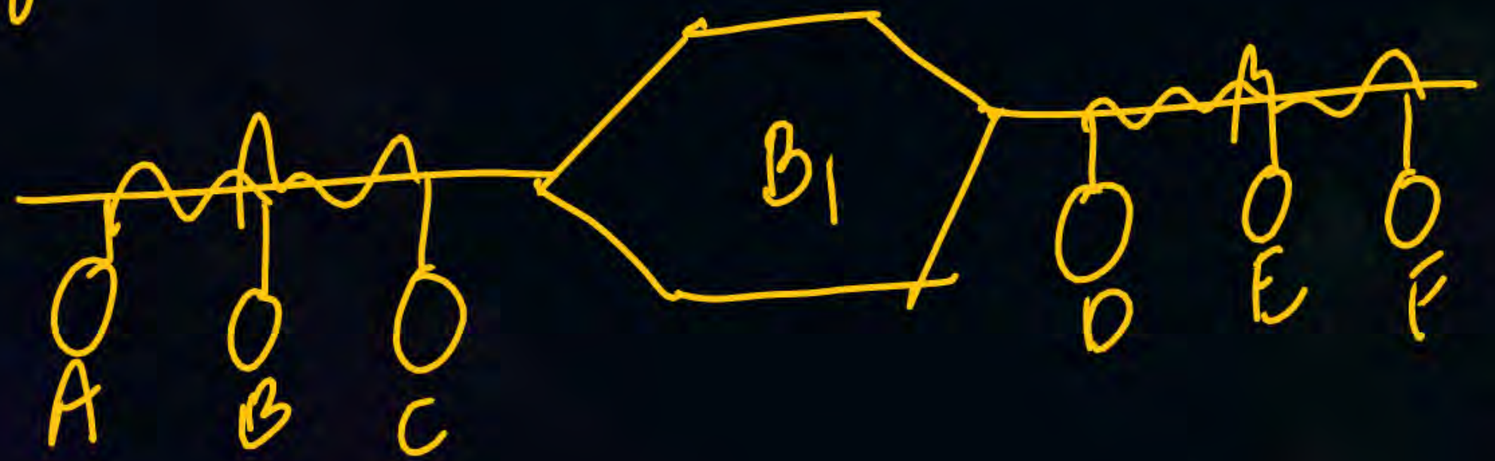
A worm hole routing



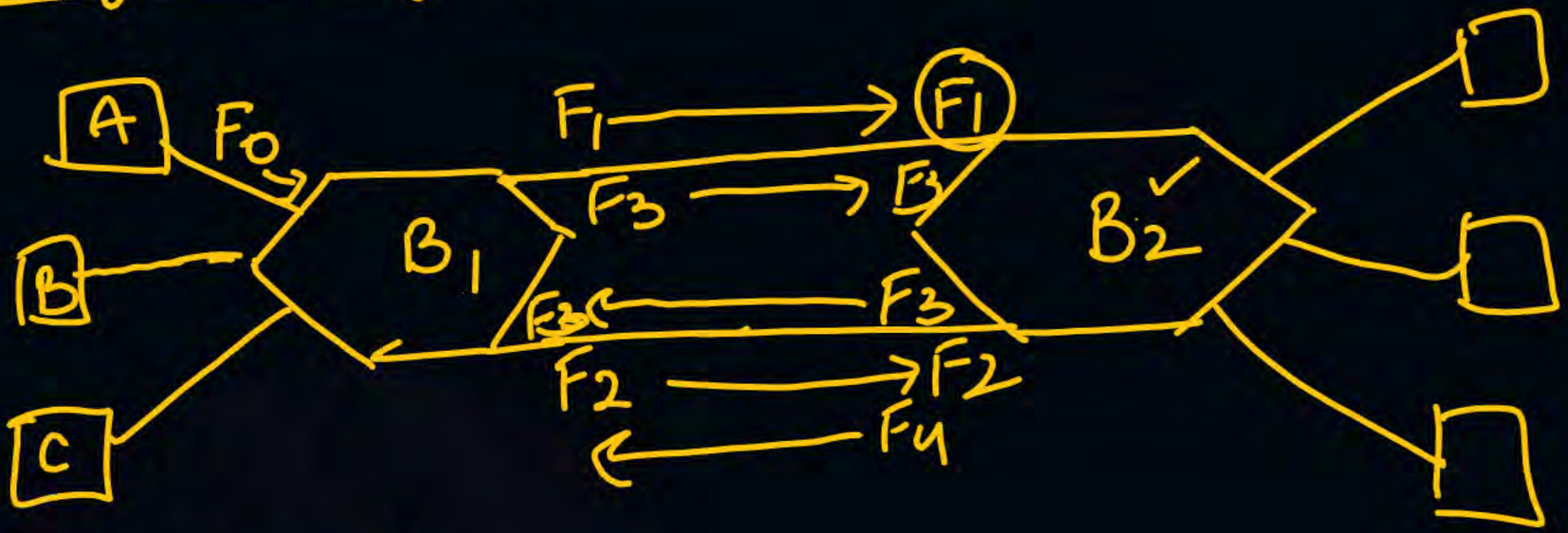

Station A
Station B
NL
DLL
PL
Packet
EH
Packet
EH
Packet
MAC
Relay
EH
P
EH
P
ETH
Packet
ETH
P
P
EH
P
EH
P
Wire

Bridge with k ports have k instances of MAC (DLL) and physical layers

* Inside bridges there will be no collision
So bridges decrease the collision domain



Spanning tree algo



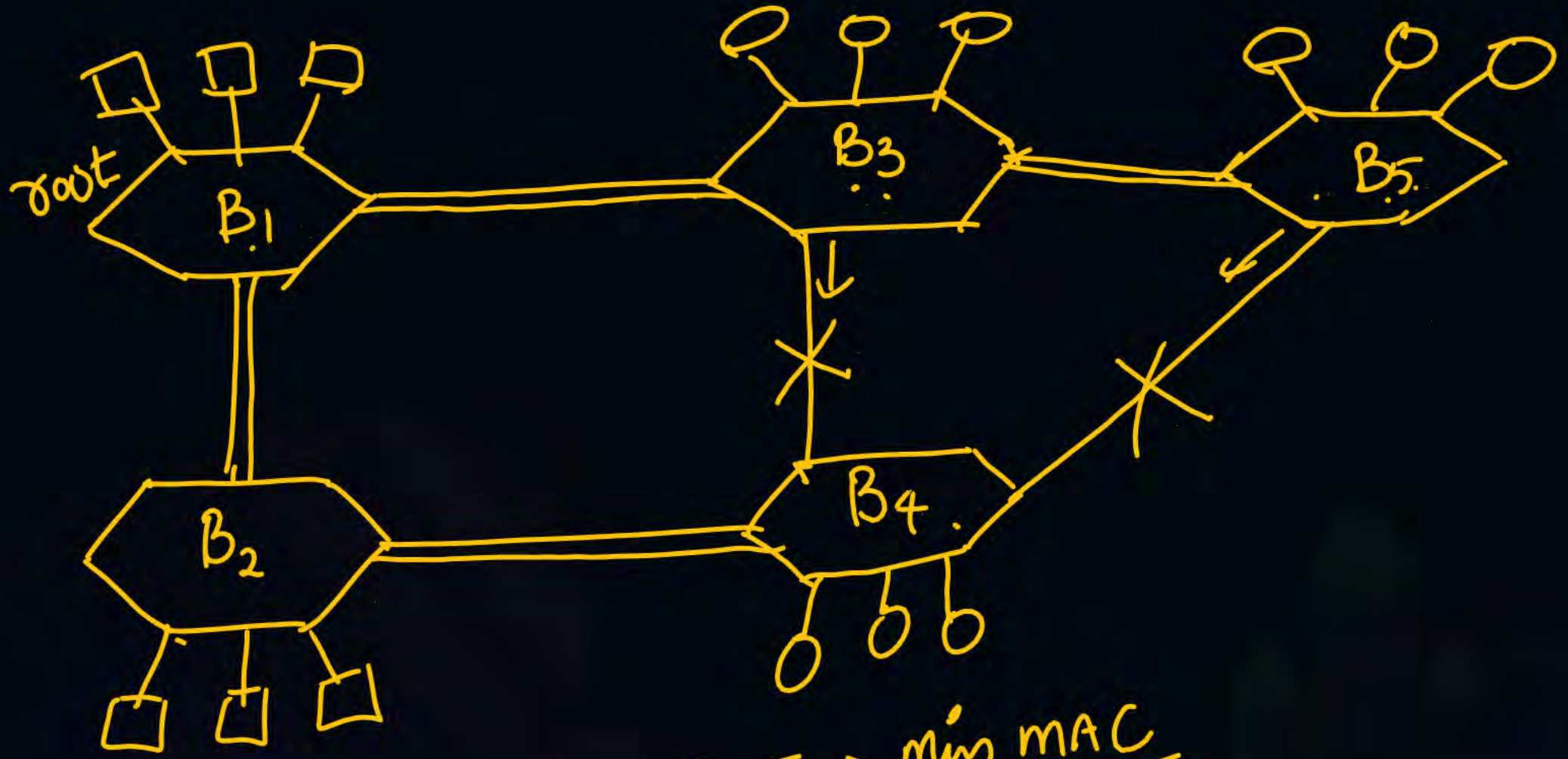
A → D

Frame falls in oo loop.

Frames does not have live

we need to break the loops → Spanning tree algo

1 week



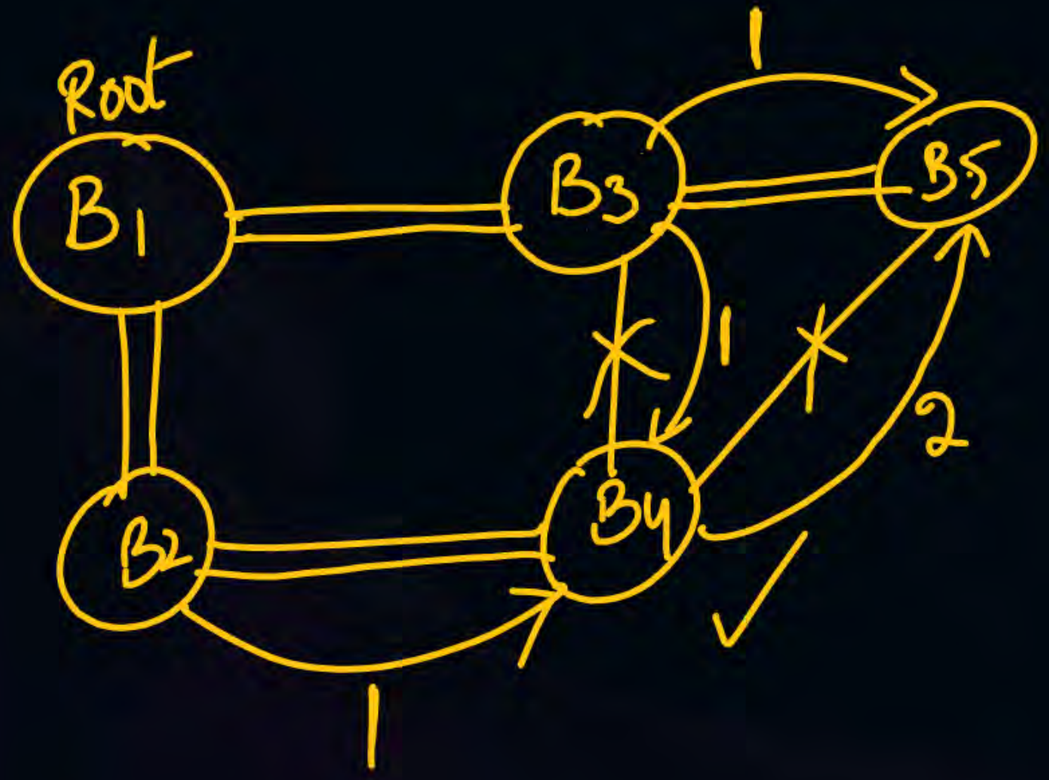
ROOT → min MAC

Bridges → MAC

→ To build a spanning tree, bridges run a distributed algo. Each bridge periodically broadcasts messages on all out going ports (neighbours) and processes the messages it receives from other bridges

→ The bridges must first choose one bridge to be the root bridge. For this they will use MAC addresses. MAC adds are installed on all bridges by the manufacturer. The lowest MAC add is the root bridge

→ Next the tree of shortest paths from the root to every bridge is constructed through configuration messages



B_2 and B_3 can reach B_1 directly in one hop

B_4 can reach B_1 through B_2 & B_3 in two hop.

But lower mac (B_2) is selected

B_5 can reach B_1 in two hops through B_3

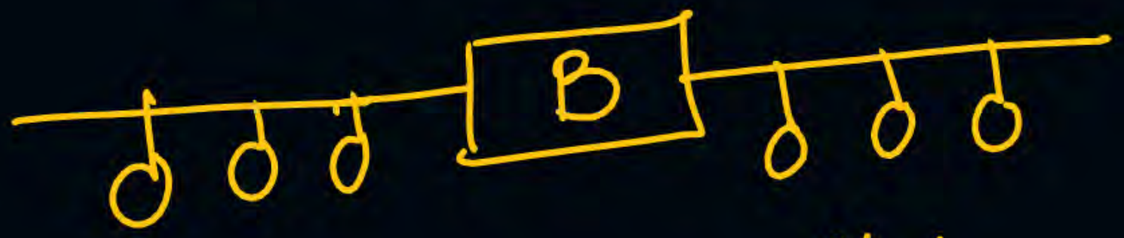
→ Each Bridge will remember the shortest paths it finds to the root. Then Bridges will turn off the ports that are not part of the shortest path

→ Even after spanning tree has been established, the algorithm continues automatically to detect topology changes and update them

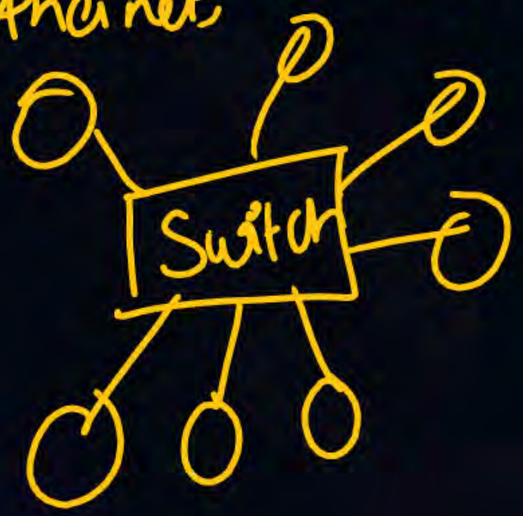
Switches:- Switches are modern Bridges by another name

Bridges were developed when classic ethernet was in use
So they tend to join relatively few LANS
Switches has more ports than Bridges and all present day
ethernets use Switches to connect the stations
Switches have PL and DLL. NO collisions are present inside a
Switch. It reduces collion domain

Classic ethernet → bus

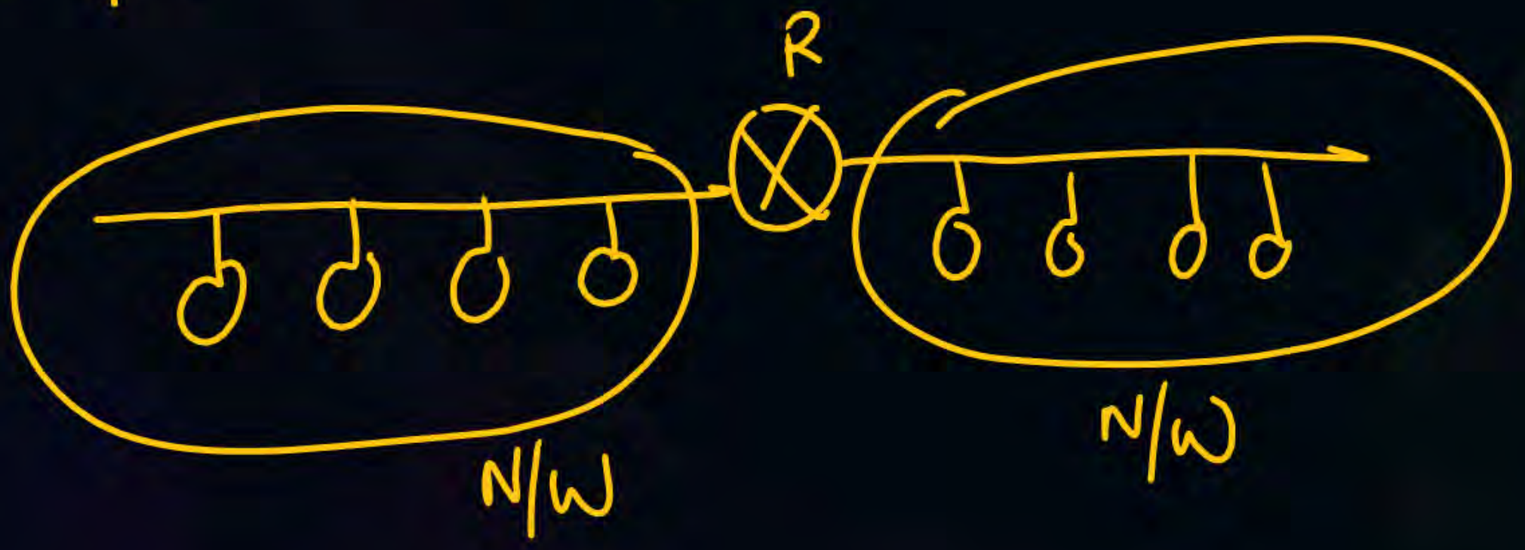


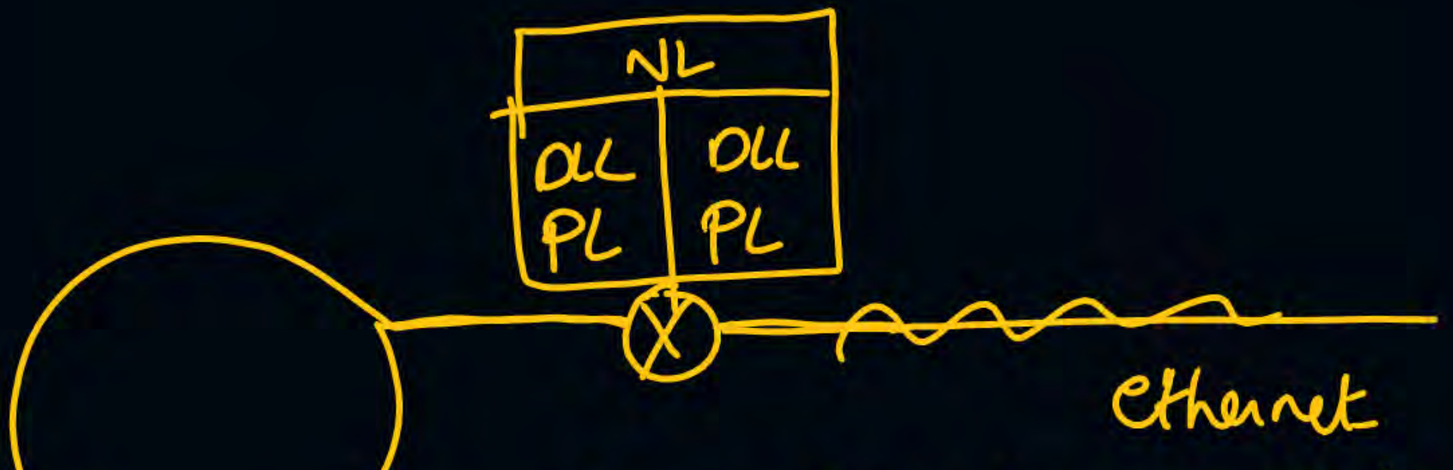
modern ethernet — Star



Routers :-

Routers are devices that connect two or more diff lans & same LANS. When router is installed, it divides one N/W into many n/w's





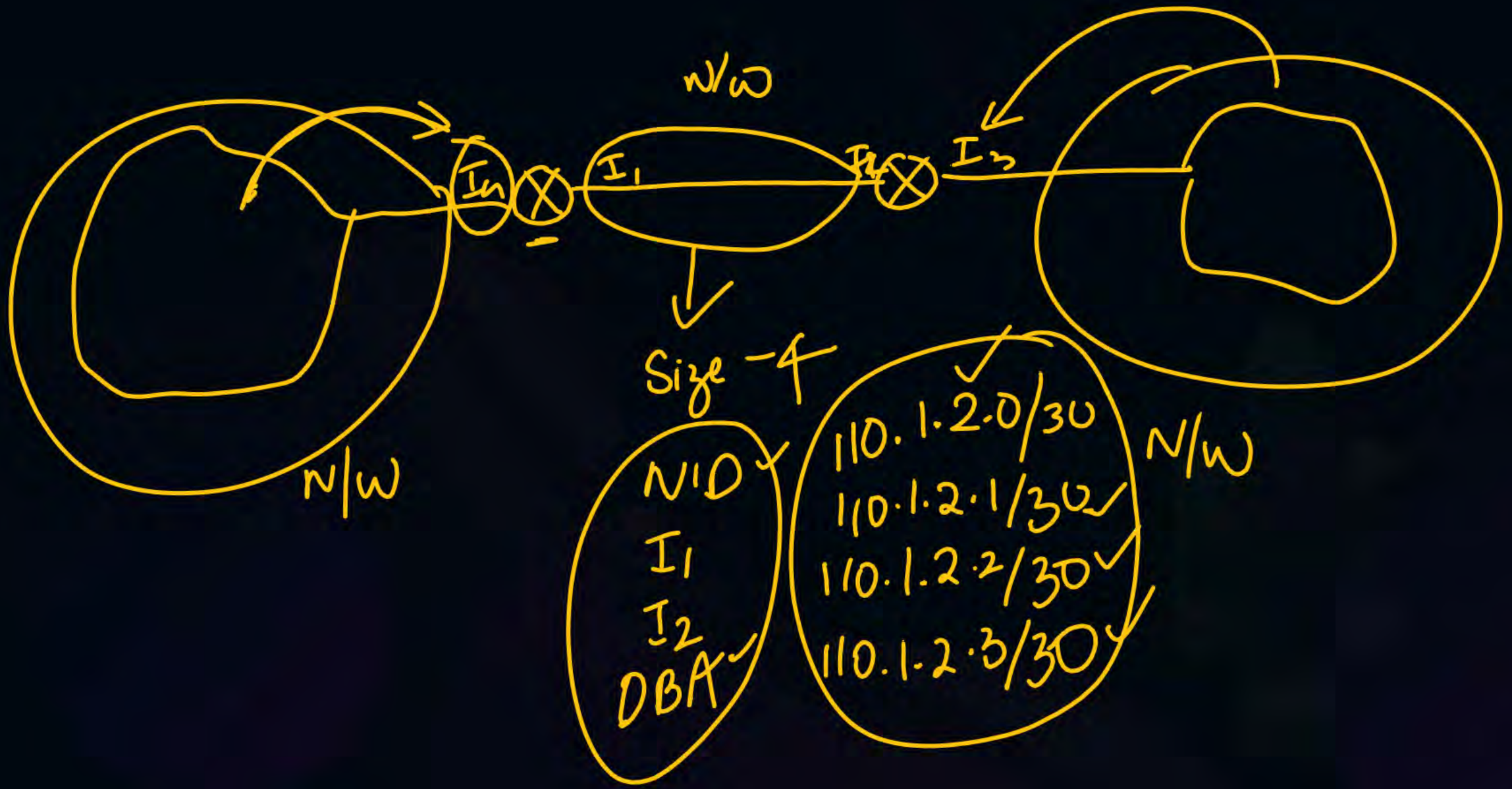
Every interface will have DLL and PL.

Every interface will have IP and MAC

(i) Filter
(ii) Forward
(iii) Flooding/Routing

(iv) NO collisions inside a router

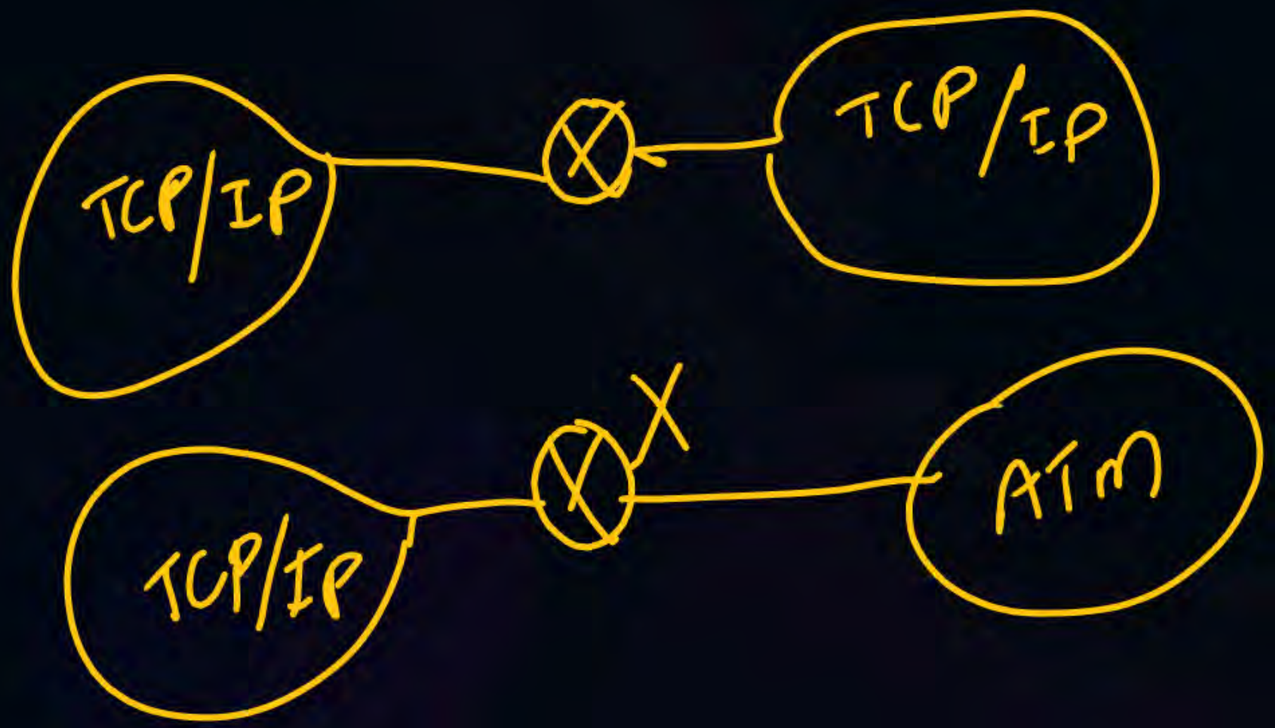
So collision domain decreases


N/W
I1
I2
I3
N/W
Size - 4
NID
I1
I2
DBA
110.1.2.0/30
110.1.2.1/30
110.1.2.2/30
110.1.2.3/30
N/W

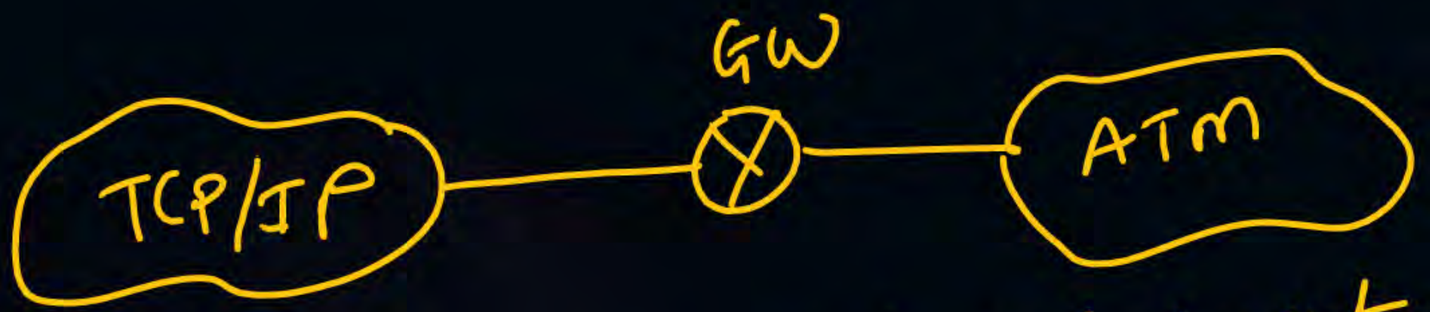
Gateways :-

Routers can connect TCP/IP N/w with TCP/IP

ATM with ATM



Gateway can connect different N/W



Gateways are used as protocol converters

Gateway can be used as a proxy



Each packet going out & coming in goes through GW and gateway can count B/w used by the users

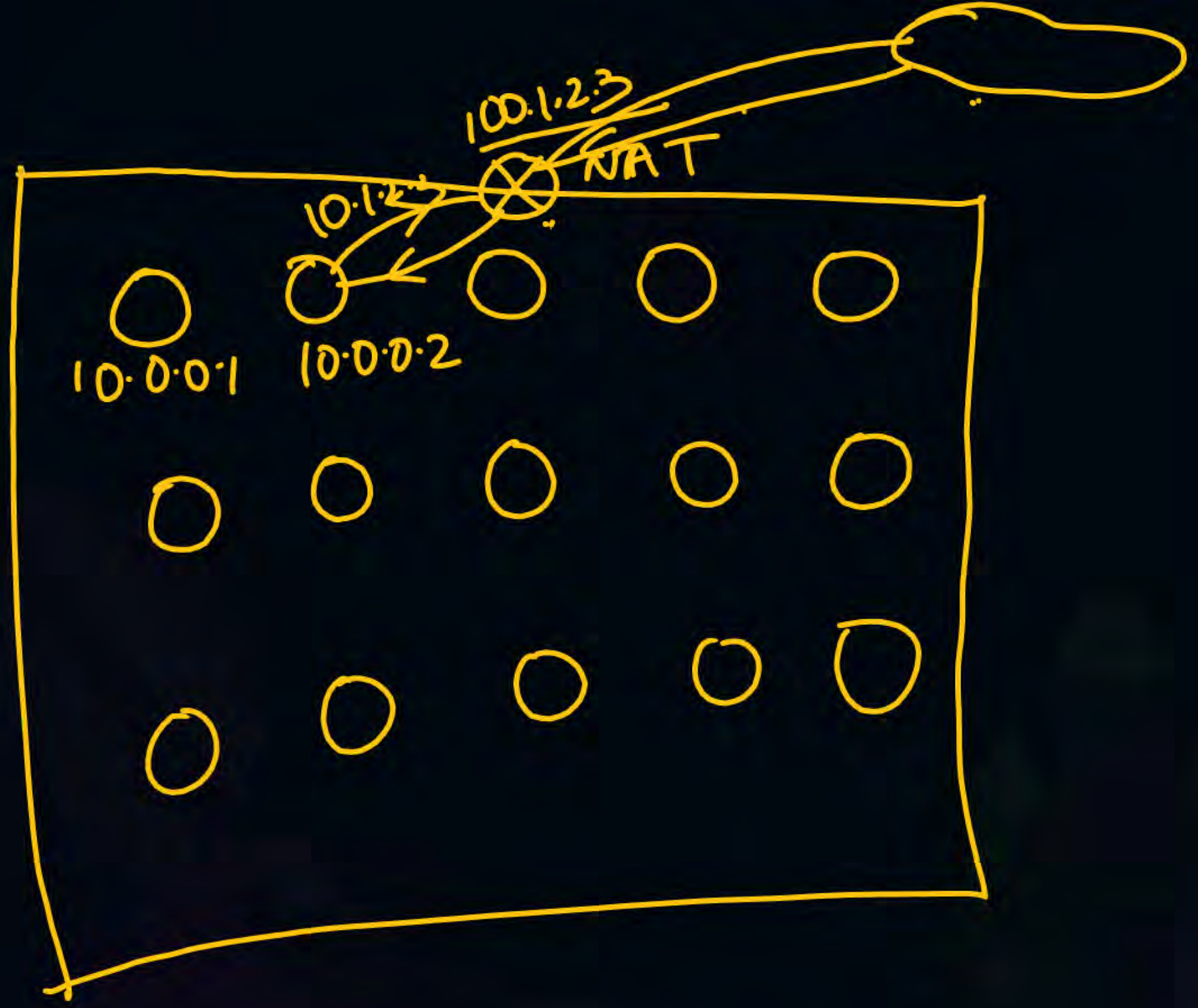
It is also used as NAT Server

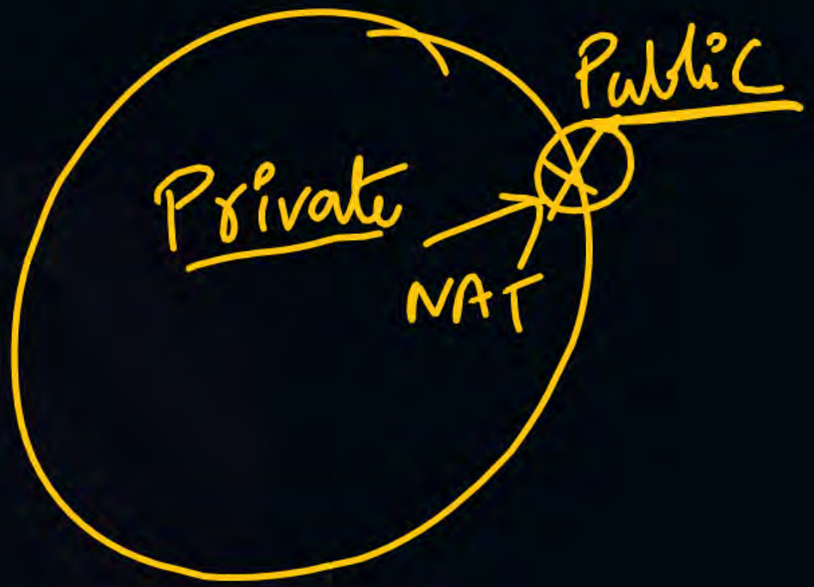
Private addresses are

10.0.0.0 to 10.255.255.255 — CA

172.16.0.0 to 172.31.255.255 — CB

192.168.0.0 to 192.168.255.255 — CC

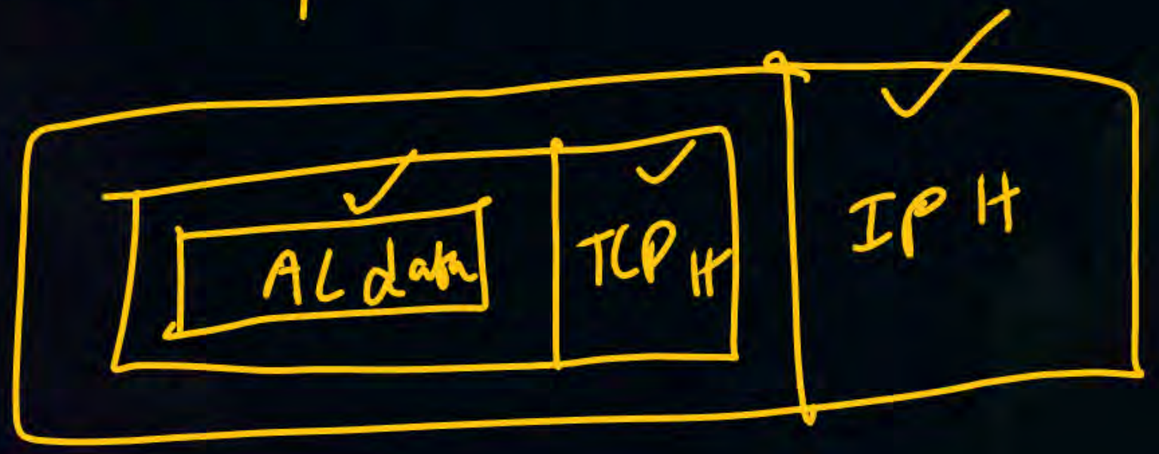

100.1.2.3
NAT
10.1.2.3
10.0.0.1
10.0.0.2


Public
Private
NAT

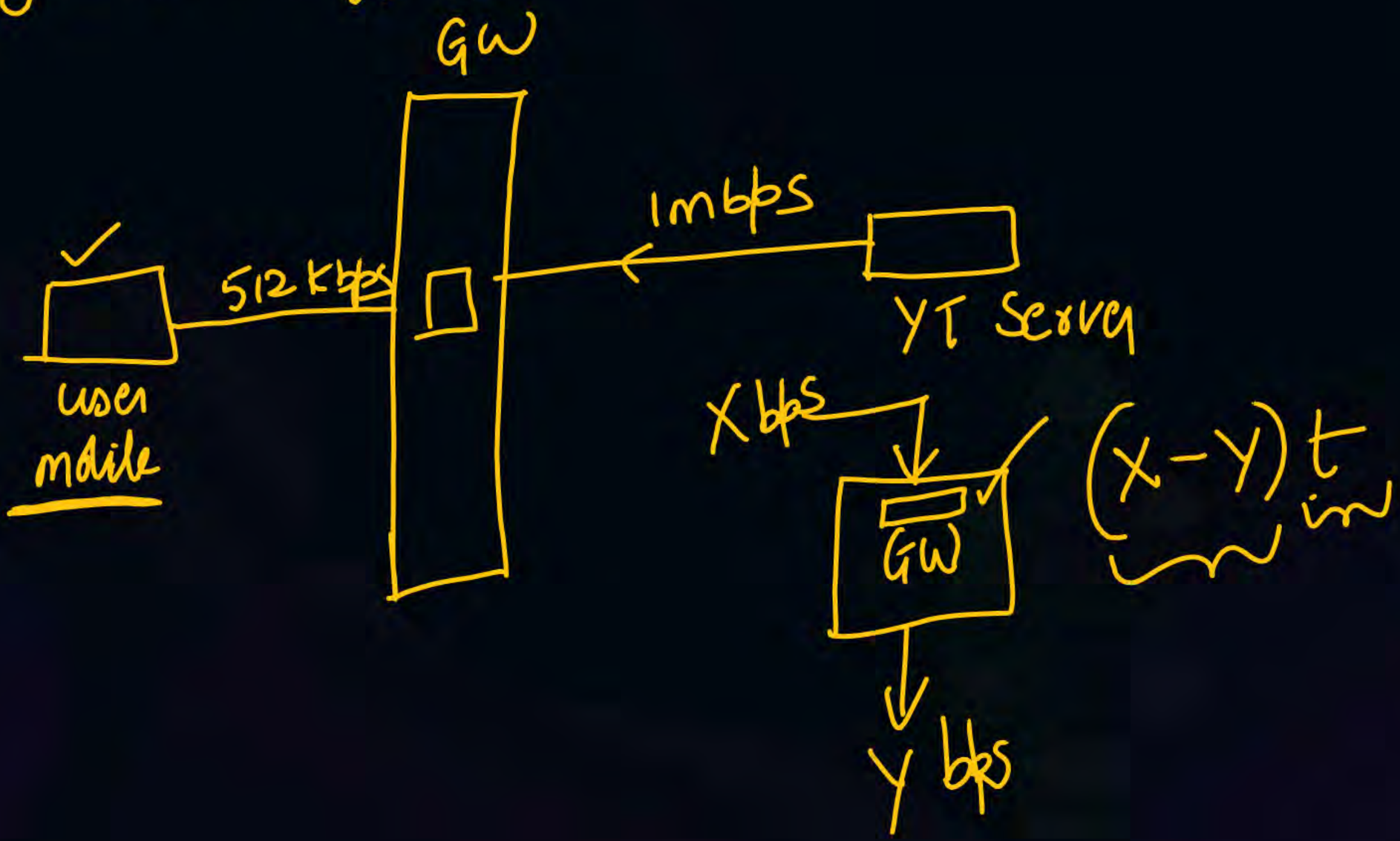
Gateway can be used as a Firewall

Gateway has all the layers. So it can look into AL Data

Deep packet inspection



Gateway can do buffering



Mobile internet GW :-

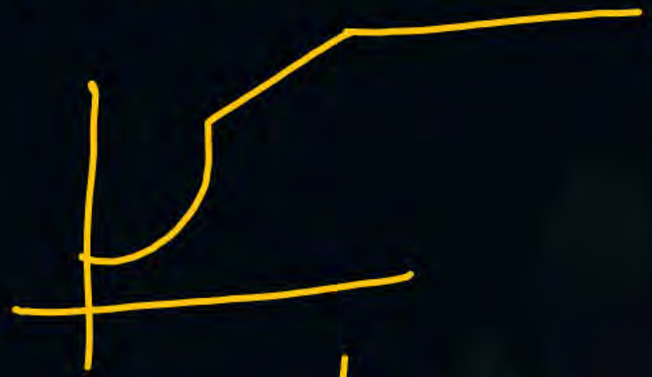


Video = 10mb

Duration = 10sec

Data Rate = $\frac{10mb}{10sec}$ = 1mbps

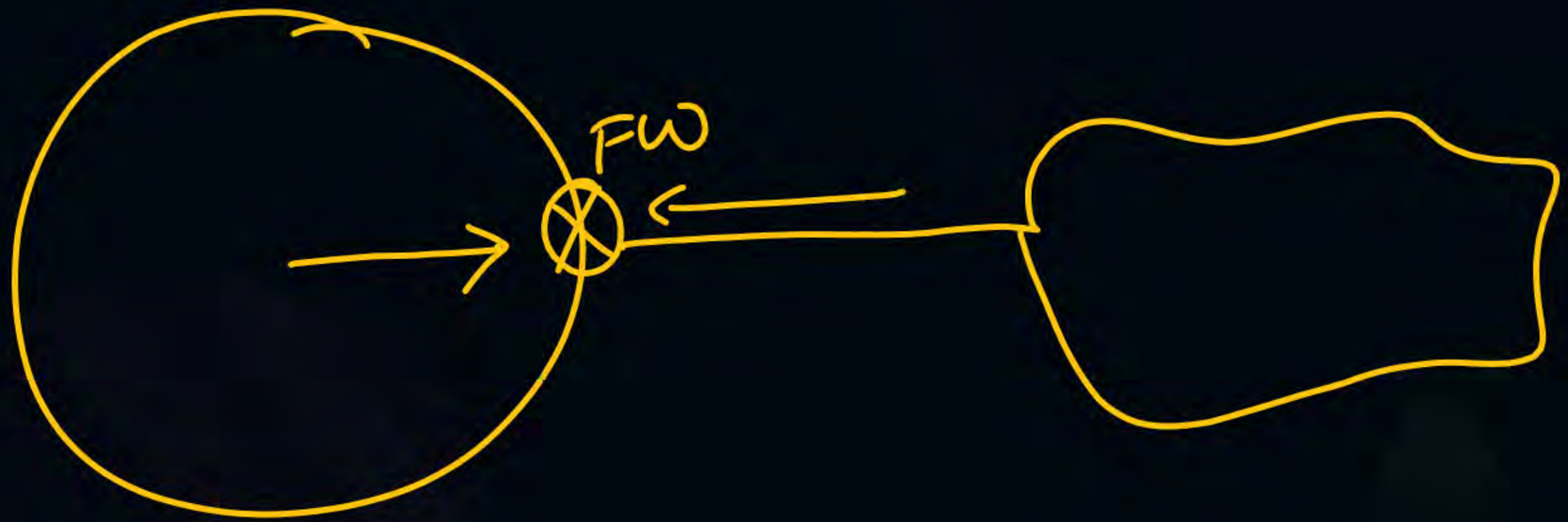
adv window flow control





DHCP by ISP is generally implemented on a GW

Firewall :-



Types of FW:-

1) layer 3 FW — PL, DLL, NL → Packet filtering Firewall
2) layer 4 FW — PL, DLL, NL, TL
3) layer 5 FW — PL, DLL, NL, AL → Proxy firewall

Layer 3 FW :- PL, DLL, NL

(i) SIP, DIP → Can block any host

(ii) Protocol field → (TCP, UDP, ICMP, IGMP)

Block any of these four protocols

(iii) Can block a particular protocol on a particular host

(I, ICMP)

Layer 4 FW :- PL, DLL, NL, TL

(i) IP (Block by IP)

(ii) Protocol (Block a protocol)

(iii) Port number (Block a service) like HTTP, SMTP, DNS

(iii) A particular servic on a partical host

(I, HTTP)

THANK - YOU